coursera

Depressive and Bipolar Disorders ✕

Today's goal • 3/3 items

Module 1
Learn With PsycLearn Essentials

Module 2
Opening to Depressive and Bipolar Disorders

Module 3
Depressive and Bipolar Disorders

Module 4
Biological Contributing Factors

Lesson 5: Genetic Contributing Factors

- ✓ Developmental Psychopathology
  Reading • 3 min
- ✓ Genetic Contributing Factors
  Reading • 3 min
- ✓ The Link Between Genes and Psychopathology
  Video • 4 min
- ✓ Practical Scenario
  Real-world application • 5 min
- ✓ Genetic Contributing Factors for Depression
  Reading • 10 min
- ✓ Epigenetic Contributors to Depression
  Video • 3 min
- ✓ Genetic Contributing Factors for Bipolar Disorder
  Reading • 3 min
- ○ Check Your Understanding: Genetic Contributing Factors
  Practice Assignment • 4 min

Lesson 6: Neurobiological Contributing Factors

- ✓ Depressive Disorders: A Case Study
  Reading • 5 min
- ✓ Communication in the Synapse
  Video • 4 min
- ✓ Neurochemical Contributors to Depression
  Reading • 10 min
- ✓ Practical Scenario
  Real-world application • 5 min
- ✓ Hormonal Contributors to Depression
  Reading • 5 min
- ✓ Brain Regions Linked to Depression
  Video • 4 min
- ○ **Neurobiological Contributors to Depressive and Bipolar Disorders**
  Reading • 15 min
- ○ Check Your Understanding: Biological Contributing Factors
  Practice Assignment • 6 min

Key Takeaways

- ○ Key Takeaways: Biological Contributing Factors
  Reading • 10 min

Module 4 Quiz

- ○ Module Quiz: Biological Contributing Factors
  Graded Assignment • 9 min

Module 5
Psychological Contributing Factors

Module 6
Treatments for Depressive and Bipolar Disorders

Module 7
Suicide

Module 8
Conclusion

Module 9
Cumulative Quiz

Module 10
Course Resources

Module 11
PsycLearn Essentials APA Student Resources

Neurobiological Contributors to Depressive and Bipolar Disorders

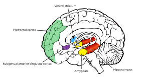
0:21 / 7:22 1x

Brain Regions and Circuits Associated with Depression

Much research has investigated brain regions involved in depression, but the research has produced conflicting findings. This situation led psychiatrist Mayur Pandya and his colleagues to conclude that identifying a single brain region as a contributor to depression is not possible because many brain regions are implicated in depression (Pandya et al., 2012; Clark et al., 2009). Recent research supports the hypothesis that depression is more likely due to dysfunction of the brain circuits that link multiple brain regions rather than any one brain region (Biselli et al., 2021). **Brain circuits** are interconnected networks of brain structures that communicate with each other (Tau & Peterson, 2010).

One way to identify brain regions that are implicated in depression is to identify symptoms of depressive disorders and link them to the brain areas that are either overactive or underactive while the person is experiencing those symptoms. Using this technique, researchers have observed atypical activity in numerous brain regions in such patients. These brain areas appear to be part of a depression-related brain circuit (Hare & Duman, 2020), which includes the prefrontal cortex, the subgenual anterior cingulate cortex, the amygdala, the ventral striatum, and the hippocampus. The image titled *Brain Structures Involved in Depressive Disorders* illustrates the location of these brain regions. These interconnected brain regions receive and process incoming signals and communicate neural messages to other parts of the circuit (Tau & Peterson, 2010).

Brain Structures Involved in Depressive Disorders



Note. The prefrontal cortex, the subgenual anterior cingulate cortex, the amygdala, and the hippocampus appear to be part of a depression-related brain circuit.

Brain Regions and Circuits Associated with Bipolar Disorder

Like many depressive disorders, bipolar disorder doesn't arise out of a single brain area but rather it is associated with dysfunctions in a large number of interconnected areas. This has led the researchers interested in understanding bipolar disorders to focus on the functioning of larger networks of brain structures and how areas within these networks connect to and communicate with each other.

There are conflicting findings about which brain structures are involved in bipolar disorder, likely owing to the variability in bipolar symptoms, such as the experience of psychotic or non-psychotic symptoms (Birur et al., 2017; Ellison-Wright & Bullmore, 2010). Regions implicated in bipolar disorder include the anterior cingulate cortex, orbitofrontal cortex, ventral striatum, hippocampus, amygdala, and insula.

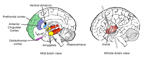
There is evidence that brain tissue loss, such as in the anterior cingulate cortex (ACC), the orbitofrontal cortex, the ventral striatum, the hippocampus, and the insula, or brain tissue gain, such as in the amygdala, are associated with bipolar disorders (Birur et al., 2017; Eker et al., 2014; Ellison-Wright & Bullmore, 2010). Other research has found decreased activity, such as in the anterior cingulate cortex and ventral striatum, which are involved with regulating emotions and with the reward circuitry in the brain that regulates feelings of pleasure (Caseras et al., 2013). Increased activity in brain areas such as the orbitofrontal cortex, amygdala, and insula, have also been linked to bipolar disorder.

This research lends evidence that bipolar disorder arises from the failure of the limbic network to sufficiently regulate the activity of the amygdala. This is consistent with findings that there is poor connectivity within the limbic network in bipolar patients relative to similar people who do not have bipolar disorder (Newman et al., 2017).

It's worth noting two points about the way the brain operates in the context of bipolar disorder. First, many of the functional changes in the activity and size of brain regions associated with bipolar disorder are not permanent. For example, researchers can tell whether a bipolar patient is depressed or not by looking at the activity of relevant areas of their brain (Newman et al., 2017). Put another way, brain activity returns to a more normal state when the bipolar episode fades. Second, recent work has suggested that structural brain changes, like the decreased volumes previously described, may be the result of the immune system causing inflammation in the brain, thereby damaging or destroying the tissue (Magioncalda & Martino, 2022).

The following figure, *Brain Structures and Bipolar Disorder*, illustrates several of the key areas that show effects of bipolar disorder. Following the figure is a table that lists the structural and functional differences arising from bipolar disorder.

Brain Structures and Bipolar Disorder



Note. Regions implicated in bipolar disorder include the anterior cingulate cortex, orbitofrontal cortex, ventral striatum, hippocampus, amygdala, and insula.

Structural and Functional Differences Arising From Bipolar Disorder

Brain structure	Function	Activity in patients with bipolar disorder	Size in patients with bipolar disorder
Anterior Cingulate Cortex (ACC)	Plays an important role in regulating and experiencing emotions	N/A	Decreased
Orbitofrontal Cortex	Involved in integrating emotions into decision making	Increased	Decreased
Ventral Striatum	Involved in the perception of pleasure and reward-related behavior	Decreased	Decreased
Hippocampus	Involved in memory creation and processing emotional stimuli	N/A	Decreased
Amygdala	Involved in the perception and processing of emotional stimuli, especially negative stimuli	Increased	Increased
Insula	Involved in negative emotions and self-awareness	Increased	Decreased

In sum, there are many different biological mechanisms that are important in understanding the causes and treatments of depressive and bipolar disorders. We know a lot about how each one operates individually and we're starting to know more and more about how they work together to create a mental disorder. The following table, *Summary of Biological Factors of Depressive and Bipolar Disorders* summarizes the biological factors covered in this module.

Summary of Biological Factors of Depressive and Bipolar Disorders

Biological factors	Nature of biological factor
Neurotransmitters	Chemicals used to send messages from one neuron to another.
Genetics	Genes provide cells the recipes for proteins needed to create neurotransmitters, receptors, and other chemicals used in the nervous system.
Epigenetics	Environmental factors can activate or deactivate genes on our DNA.
Hormones	Hormones are chemical messengers used by the body to communicate messages from one location to another.
Brain Circuits	Although researchers initially looked at individual brain structures and areas, they now talk about groups of interconnected brain areas involved in depression.

Mark as completed

Like Dislike Report an issue

Go to next item →